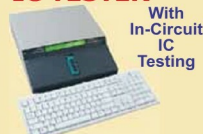


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Q. What's the difference between Ethernet, Wi-Fi and WiDi, and how do these work?

Siva Pramod
Through e-mail

A. Ethernet, Wi-Fi and WiDi are networking solutions for linking computers and peripherals, employing different media and protocols.

The Ethernet. In 1973, researcher Bob Metcalfe designed and tested the first Ethernet network connecting a computer to a printer with a cable and operating with a protocol that governed communication on the cable. Ethernet has since become the most popular and widely deployed network technology in the world. The Ethernet standard has grown to encompass new technologies as computer networking matured.

Originally, the Ethernet entailed communication over a single cable shared by all devices on the network. Once a device was attached to this cable, it had the ability to communicate

tify the intended recipient of the frame.

For example, in Fig. 1, when computer B transmits to printer C, computers A and D will still receive and examine the frame. However, when a station first receives a frame, it checks the destination address to see whether the frame is intended for it. If it is not, the station discards the frame without even examining its contents.

Electrical signals propagate along a cable very quickly, but weaken as they travel due to electrical interference from neighbouring devices. A network cable must be short enough so that devices at opposite ends can receive signals from each other clearly and with minimal delay. This places a distance limitation on the maximum separation between two devices (called the network diameter) on an Ethernet network. Additionally, since in carrier-sense multiple-access/collision detection only a single device can transmit at a given time, there are practical limits to the number of devices that can

coexist in a single network. If too many devices are attached to one shared segment, the contention for the medium will increase and every device may have to wait for an inordinately long time before getting a chance to transmit.

A number of devices have been developed that alleviate these problems. Many of these devices are not specific to Ethernet, but play role in other network technologies.

Wi-Fi. Wi-Fi or 802 is wireless networking of computers to provide free or low-cost Internet. It has many advantages over Ethernet. Wireless networks are easy to set up and inexpensive. These are also unobtrusive. A Wi-Fi network uses radio waves, whereas Ethernet uses physical medium (cable).

Wi-Fi is a lot like two-way radio communication. A computer's wireless adaptor translates data into a radio signal and transmits it using an antenna. A wireless router receives the signal and decodes it. The router sends

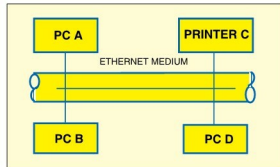


Fig. 1: PC and printer networked on Ethernet

with it. This allowed the network to expand to accommodate new devices without requiring any modification to the already existing devices on the network.

Today, Ethernet is a local-area technology, with networks traditionally operating within a single building, connecting devices in close proximity. Technological advancements have increased these distances considerably, allowing Ethernet networks to span over kilometres.

Since a signal on the Ethernet medium reaches every attached node, the destination address is critical to iden-